GYUMRI, Armenia

WMO: **376860**

Lat: **40.763N** Lon: **43.856E** Elev: **1513** StdP: **84.42** Time Zone: **4.00 (E04)** Period: **07-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-23.1	-20.9	-26.0	0.4	-23.3	-23.2	0.6	-20.5	9.5	-3.9	8.0	-3.0	0.9	40	0.600	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	14.3	31.9	16.4	30.1	16.4	28.8	16.0	18.2	27.7	17.5	26.9	16.7	25.9	4.0	70	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	15.0	12.8	23.3	14.1	12.0	21.6	13.2	11.3	20.8	58.3	27.6	55.6	27.0	53.3	25.8	21.0	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	11.0	9.6	8.3	DB	-23.8	34.2	5.2	2.0	-27.6	35.6	-30.6	36.8	-33.6	38.0	-37.4	39.4	(4)
(5)				WB	-24.2	19.5	5.1	0.9	-27.8	20.1	-30.8	20.6	-33.7	21.1	-37.4	21.7	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	7.8	-7.5	-5.6	1.6	7.8	12.2	17.0	21.1	21.5	16.5	9.8	2.4	-4.2	(6)
(7)			DBStd	10.68	5.82	6.17	4.77	3.39	2.47	2.82	2.74	2.85	3.67	3.02	3.83	5.45	(7)
(8)			HDD10.0	2039	542	436	261	82	8	0	0	0	3	40	228	440	(8)
(9)			HDD18.3	4081	800	669	518	316	189	58	8	7	75	263	478	698	(9)
(10)			CDD10.0	1236	0	0	1	17	77	209	343	356	197	36	0	0	(10)
(11)			CDD18.3	234	0	0	0	0	0	18	93	104	19	0	0	0	(11)
(12)			CDH23.3	2527	0	0	0	4	7	244	927	1022	318	5	0	0	(12)
(13)			CDH26.7	818	0	0	0	0	0	41	314	382	81	0	0	0	(13)
(14)	Wind		WSAvg	2.4	1.2	1.2	2.5	2.7	2.4	2.9	4.1	4.3	2.7	1.8	1.4	1.1	(14)
(15)	Precipitation		PrecAvg	471	22	25	29	55	82	70	46	32	26	37	26	24	(15)
(16)			PrecMax	793	47	62	71	123	137	162	109	82	72	112	57	54	(16)
(17)			PrecMin	348	1	3	0	4	30	22	5	0	0	4	3	1	(17)
(18)			PrecStd	89	12	14	16	25	27	36	29	19	19	24	14	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		5.2	9.2	17.1	22.9	24.2	30.0	33.9	34.8	31.2	24.0	15.2	10.8	(19)
(20)			MCWB		2.2	4.7	7.9	12.2	13.3	15.3	16.0	16.2	15.0	11.0	7.2	3.7	(20)
(21)		2%	DB		3.0	6.2	14.0	19.9	22.2	27.9	31.9	32.1	29.0	21.9	13.8	6.2	(21)
(22)			MCWB		0.6	2.8	6.6	10.0	12.5	15.3	17.2	16.1	14.1	10.8	6.5	1.9	(22)
(23)		5%	DB		1.9	4.1	11.8	17.9	20.9	26.0	30.0	30.2	26.9	19.8	11.2	4.1	(23)
(24)			MCWB		0.0	1.3	5.3	8.9	12.2	15.1	17.3	16.3	13.9	10.4	5.7	1.7	(24)
(25)		10%	DB		0.8	2.1	9.5	15.8	19.0	24.2	28.1	28.9	25.0	17.2	9.2	2.8	(25)
(26)			MCWB		-0.4	0.2	4.3	7.9	11.3	14.8	16.7	16.1	13.3	9.7	4.8	0.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		3.1	5.2	9.0	13.5	15.0	17.9	19.8	19.3	16.7	12.7	8.7	4.7	(27)
(28)			MCDB		4.0	8.9	15.2	22.3	21.7	26.0	29.7	29.7	26.8	19.3	13.0	7.4	(28)
(29)		2%	WB		1.1	3.0	7.1	10.7	13.8	16.9	18.7	17.9	15.5	11.9	7.2	3.0	(29)
(30)			MCDB		1.9	5.7	13.0	17.7	20.0	25.2	28.3	28.1	25.6	18.7	11.0	5.1	(30)
(31)		5%	WB		0.2	1.5	5.8	9.5	12.9	16.1	17.9	17.1	14.7	11.1	6.4	1.7	(31)
(32)			MCDB		1.3	3.5	10.5	15.8	18.9	23.9	27.0	27.2	24.4	17.4	10.3	3.3	(32)
(33)		10%	WB		-0.8	0.3	4.6	8.5	11.9	15.3	17.2	16.4	14.0	10.3	5.5	0.7	(33)
(34)			MCDB		0.5	1.8	8.6	14.4	17.7	22.8	26.2	26.2	23.0	16.2	9.0	2.1	(34)
(35)	Mean Daily Temperature Range		MDBR		10.0	11.3	11.9	13.2	12.7	14.0	14.0	14.3	15.5	13.8	12.7	10.1	(35)
(36)		5% DB	MCDBR		8.8	10.8	15.3	16.7	15.6	15.8	15.9	16.0	17.7	18.1	16.0	12.1	(36)
(37)			MCWBR		6.7	7.6	8.6	8.2	7.7	6.5	5.6	5.2	6.2	8.8	10.0	8.6	(37)
(38)		5% WB	MCDBR		6.8	9.7	13.1	14.6	13.6	14.2	14.2	14.4	15.9	14.9	14.0	9.8	(38)
(39)			MCWBR		5.4	6.9	7.7	7.4	7.1	6.5	5.5	5.2	5.9	7.6	8.9	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.275	0.307	0.356	0.413	0.388	0.371	0.407	0.400	0.345	0.342	0.294	0.272	(40)
(41)		taud			2.336	2.254	2.202	2.098	2.259	2.352	2.265	2.281	2.418	2.370	2.463	2.414	(41)
(42)		Ebn at Noon			886	901	888	857	888	902	865	862	894	851	862	865	(42)
(43)		Edh at Noon			85	109	129	154	134	123	133	127	103	97	76	73	(43)
(44)	All-Sky Solar Radiation	RadAvg			2.33	3.45	4.23	4.70	5.54	6.67	6.74	6.18	5.00	3.28	2.22	1.76	(44)
(45)		RadStd			0.21	0.36	0.44	0.50	0.36	0.43	0.38	0.35	0.34	0.27	0.29	0.23	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page